PCIE INTERVIEW QUESTIONS

❖ **VERILOG PROJECT NAME: PERIPHERAL COMPONENT INTERCONNECT EXPRESS (PCIe)**

1. What is a PCIE?

A PCIE is a peripheral component interconnect express. It is a high-speed serial computer expansion bus standard.

2. Can you explain the different generations of PCI Express?

The first generation of PCI Express was released in 2004 and operated at 2.5 GT/s. The second generation was released in 2007 and operated at 5 GT/s. The third generation was released in 2010 and operated at 8 GT/s. The fourth generation was released in 2017 and operated at 16 GT/s.

3. What are some typical applications that use PCI Express?

Some typical applications that use PCI Express include:

–Servers
–Workstations
–PCs
–Laptops
–Tablets
–Smartphones
–Embedded systems
–Networking equipment
–Storage devices
–Graphics cards
– Peripherals

4. What’s the difference between PCIe and PCI-X?

The main difference between PCIe and PCI-X is that PCIe is a serial bus while PCI-X is a parallel bus. This means that PCIe can transmit data much faster than PCI-X, but it also uses more power.

5. What do you understand about ACK/NACK signals in PCI Express? How are they used?

ACK/NACK signals are used in PCI Express to indicate whether or not a particular transaction was successful. If the transaction was successful, then the receiver will send an ACK signal. If the transaction was not successful, then the receiver will send a NACK signal.

6. Is it possible to run two devices on the same lane in PCI Express? If yes, how would you go about doing so?

Yes, it is possible to run two devices on the same lane in PCI Express. You would need to configure the devices to use the same lane, and then you would need to configure the PCI Express controller to allow for multiple devices on the same lane.

7. What happens when there is a conflict during the arbitration phase in PCI Express?

When two devices on a PCI Express bus try to initiate a transaction at the same time, they must go through an arbitration process in order to determine which one gets to go first. This process is known as the arbitration phase. If there is a conflict during this phase, then the device with the higher priority will be given priority and will be able to initiate its transaction first.

8. Do all lanes support the same bandwidth in PCI Express? If not, then why not?

No, all lanes in PCI Express do not support the same bandwidth. The reason for this is that the data transfer rate is determined by the number of lanes that are being used. A lane is a pair of differential signaling wires, so a x1 connection uses one lane, a x4 connection uses four lanes, and so on. The more lanes that are used, the higher the data transfer rate will be.

9. Can you explain what PIPE stands for in context with PCI Express?

PIPE is an acronym for PCI Express Interconnect Protocol. It is the specification that defines how data is exchanged between two PCI Express devices.

10. Why are endpoints needed in PCI Express? What role do they play?

Endpoints are needed in PCI Express in order to provide a connection between the root complex and the various devices that are attached to the bus. They act as a go-between, translating requests and commands from the root complex into a format that the attached devices can understand.

11. What are some examples of real-world products that use PCI Express?

Many computer motherboards and expansion cards use PCI Express, as do some routers and storage devices.

12. What do you understand about virtual channels in PCI Express?

Virtual channels are a feature of PCI Express that allows multiple logical channels to be multiplexed onto a single physical channel. This allows for increased bandwidth and improved efficiency.

13. What is the usage of split transactions in PCI Express?

Split transactions are used in PCI Express to allow for data to be split up into multiple transactions in order to improve performance. This is especially useful when dealing with large amounts of data that need to be transferred quickly.

14. What are posted transactions in PCI Express?

Posted transactions are transactions that are initiated by the device, rather than the host. They do not require a response from the host, and they are typically used for things like interrupts or other notifications.

15. What is your understanding of atomic operations in PCI Express?

Atomic operations in PCI Express are used to ensure that data is transferred between two devices in a safe and reliable manner. These operations are typically used when data needs to be transferred between two devices that are not connected directly to each other.

16. What is hot plugging? How is it implemented in PCI Express?

Hot plugging is the ability to add or remove devices from a system while it is still running. This is implemented in PCI Express through the use of hot plug controllers (HPCs). These controllers manage the power and configuration settings for devices that are plugged into the system, and they can also perform error checking to make sure that the devices are functioning properly.

17. How can data be transferred over multiple lanes in PCI Express?

Data can be transferred over multiple lanes in PCI Express by using a technique called lane aggregation. This allows for multiple lanes to be combined in order to increase the bandwidth of the connection.

18. What are the main differences between PCI Express and USB 3.0? Which one would you recommend using in certain situations?

The main difference between PCI Express and USB 3.0 is that PCI Express uses a point-to-point topology while USB 3.0 uses a hierarchical topology. This means that each device on a PCI Express bus has its own dedicated connection to the root complex, while USB 3.0 devices are connected in a daisy-chain fashion. In terms of performance, PCI Express is typically faster than USB 3.0, although this will vary depending on the specific implementation. In terms of power consumption, USB 3.0 is typically more power-efficient than PCI Express.

In terms of which one to use in certain situations, it really depends on the specific needs of the application. If speed is the most important factor, then PCI Express is the better choice. If power consumption is more important, then USB 3.0 is the better choice.

19. Is PCIe better than SSD?

PCIe or Peripheral Component Interconnect Express SSDs are more desirable and expensive than SATA SSDs. PCIe SSDs have a more direct connection to your system's motherboard. It is commonly used with devices that need extremely fast data connections — like a graphics card.

20. What is PCIe ?

PCIe protocol training is a 6 weeks course(weekends training). It covers all the aspects of PCIe Gen1 to Gen4, including PCIe topology, configuration headers, enumeration, Transaction layer, Data link layer, Physical layer, reset, power management, interrupt handling, error handling.

21. What is PCS in PCIe?

PCS is the sublayer of the physical layer of PCI Express 1.0. The major constituents of this layer are transmitter and receiver. Transmitter comprises of 8b/10b encoder. The Primary purpose of this scheme is to embed a clock into the serial bit stream of transmitter lanes.

22. What is PCS and PMA in PCIe?

The physical coding sublayer (PCS) is a networking protocol sublayer in the Fast Ethernet, Gigabit Ethernet, and 10 Gigabit Ethernet standards. It resides at the top of the physical layer (PHY), and provides an interface between the Physical Medium Attachment (PMA) sublayer and the media-independent interface (MII)

23. What are the layers in PCIe?

The PCI Express protocol consists of a Transaction Layer, a Link Layer, and a Physical Layer. Within each layer, there are corresponding sublayers. The Link Layer consists of the Media Access Control Sublayer. The Physical Layer consists of the Logical and Electrical Sublayers.

24. Why scrambler is used in PCIe?

Data Scrambling - PCI Express employs a technique called data scrambling to reduce the possibility of electrical resonances on the link. PCI Express specification defines a scrambling/descrambling algorithm that is implemented using a linear feedback shift register.

25. What is difference between scrambling and encryption?

Even if the scrambling uses some kind of key, it's a simple one that can be guessed by trying all of the possible values. Encryption, on the other hand, uses mathematics to ensure that both sides must have corresponding keys in order to decrypt the data, even if the exact algorithm or “puzzle” is known

26. What is PCIe architecture?

Peripheral Component Interconnect Express (PCIe or PCI-E) is a serial expansion bus standard for connecting a computer to one or more peripheral devices. PCIe provides lower latency and higher data transfer rates than parallel busses such as PCI and PCI-X.

27. What is the difference between Link and Lane in PCIe?

PCIe link and lane: A PCI Express Link is the physical connection between two devices. A Lane consists of signal pairs in each direction. An x1 Link consists of 1 Lane or 1 differential signal pair in each direction for a total of 4 signals.

28. What is a lane in PCIe?

PCIe lanes are the physical link between the PCIe-supported device and the processor/chipset. PCIe lanes consist of two pairs of copper wires, typically known as traces that run through the motherboard PCB, connecting the PCIe-enabled device to either the processor or motherboard chipset.

29. What is GT S vs GBps?

The latest PCI Express standard, PCIe 5, represents a doubling of speed over the PCIe 4.0 specifications. We're talking about 32 Giga transfers per second (GT/s) vs. 16GT/s, with an aggregate x16 link bandwidth of almost 128 Gigabytes per second (GBps)

30. Why PCIE is a serial protocol Why not parallel?

Overall, parallel is a thing of the past. Reason: too many wires, too many connections, too much interference. Bus width grows all the time and in order to support 64-bit transfers on higher speeds it would require more engineering efforts to be made (PCI-X does that).

31. Why do we use serial instead of parallel?

Serial mode offers the advantage of fewer traces on the pc board, and fewer pins on the devices. Parallel offers the advantage of transferring 8 data bits per

I/O clock cycle, but at the disadvantage of many more pins required on the devices

32. Is PCIe a serial or parallel bus?

PCIe is a serial interface whereas PCI is a parallel interface.

33. What is pcs in serdes?

The SERDES is primarily comprised of the Physical Medium Dependent (PMD) sublayer, the Physical Media Attachment (PMA) sublayer and the Physical Coding Sublayer (PCS). The PMD is the electrical block responsible for the serial signal transmission

34. What are ECRC and LCRC?

The LCRC, or Link CRC, is handled in the data link layer. The optional ECRC, or endpoint CRC, is handled in the transaction layer. Each of these CRC functions can be used to verify that correct data has been received, but they differ on handling the error. The LCRC is used to verify data sent across a single PCIe link

35. Why 8b/10b encoding is required?

The 8B/10B encoding serves two purposes. First, it makes sure there are enough transitions in the serial data stream so the clock can be recovered easily from the embedded data. Second, because it transmits the same number of ones as zeros, it maintains a d-c balance.

36. Why do we use scrambling in PCIe?

How does scrambling impact the PCIe 3.0 architecture? Scrambling is a technique where a known binary polynomial is applied to a data stream in a feedback topology. Because the scrambling polynomial is known, the data can be recovered by running it through a feedback topology using the inverse polynomial.

37. Why is scrambling used in LTE?

By using the scrambling code, Node can separate signals coming simultaneously from many different UEs and UE can separate signals coming simultaneously from many different Node.

38. What is disparity in PCIe?

The data symbols are the encoded information. These symbols (data and control) can have either a positive, negative, or zero disparity—that is, disparity defines the count difference between the number of 1s and 0s in the symbol. By coding rule, the disparity is limited to +1,-1, or zero.

39.What is Ltssm in PCIe?
The LTSSM (Link Training and Status State Machine) block checks and memorizes what is received on each lane, determines what should be transmitted on each lane and transitions from one state to another.

40. What are the four types of PCIe slots?

PCIe slots come in different physical configurations: x1, x4, x8, x16, x32. The number after the x tells you how many lanes (how data travels to and from the PCIe card) that PCIe slot has.

41. How many risers are in PCIe?

Most motherboards only have one or two PCIe X16 slots for installing video cards. This is why miners use PCIe riser cards. However, even then most boards only have four to six PCIe slots available. Well it turns out there's another solution: 4 in 1 PCIe risers.

42. What is PCIe endpoint mode?

The Layers cape Endpoint mode driver is developed based on the Endpoint framework to create endpoint controller driver, endpoint function driver, and use configfs interface to bind the function driver to the controller driver.

43. What is the frequency of PCIe?

The PCIe standard specifies a 100 MHz clock (Refclk) with at least ±300 ppm frequency stability for Gen 1, 2, 3 and 4, and at least ±100 ppm frequency stability for Gen 5, at both the transmitting and receiving devices.

44. What is root space in PCI?

The PCIE Root Complex holds a master copy of a 'Type 1 Configuration Table' that defines the host memory space that is accessible from each End
Point device.

45. What is link equalization?

Link Equalization involves a precisely timed dynamic negotiation between an Upstream Port and Downstream Port to optimally tune both transmitter (Tx) and receiver (Rx) equalization filters so that the link will operate at a BER of 10-12– 1 bit error in 1,000,000,000,000 bits received – or better.

46. What are Ltssm States?

The LTSSM consists of 11 top-level states: Detect, Polling, Configuration, Recovery, L0, L0s, L1, L2, Hot Reset, Loopback, and Disable. These states can be grouped into five categories: The Link Training states.

47. What are the PCIe layers?

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48. What is Lane polarity in PCIe?

PCI Express uses a differential signaling technology and the two signal wires are designated positive and negative (D+ and D-)

49. Which PCIe slot is fastest?

PCIe 4.0 is twice as fast as PCIe 3.0. PCIe 4.0 has a 16 GT/s data rate, compared to its predecessor's 8 GT/s. In addition, each PCIe 4.0 lane configuration supports double the bandwidth of PCIe 3.0, maxing out at 32 GB/s in a 16-lane slot, or 64 GB/s with bidirectional travel considered.

50. What is PCIe dual mode?

The “complex endpoint” chip include two independent PCI Express links. The “endpoint / root” chip includes an additional type of digital PCI Express IP, DM (dual mode) which can operate as a root port, as shown here. It can also operate as an endpoint, for example, when plugged into a PC motherboard slot.

51. How does PCIe transmit data?

With PCIe, data is transferred over two signal pairs: two wires for transmitting and two wires for receiving. Each set of signal pairs is called a "lane," and each lane is capable of sending and receiving eight-bit data packets simultaneously between two points.

52. What is the difference between PCIe switch and bridge?

A Bridge has only 2 ports. A switch can handle many ports. A Bridge is a device that connects two LANs and controls data flow between them. A Switch is a networking device that learns which machine is connected to its port by using the device's IP Address.

53. What is upstream and downstream in PCIe?

PCI Express–Upstream/Downstream

▪ An Upstream Port is a port that points in the direction of the root complex. ▪ A Downstream Port is a port that points away from the root complex.

54.What is PCIe Equalization?

Equalization is a method of distorting the data signal with a transform representing an approximate inverse of the channel response. It may be applied either at the Tx, the Rx, or both. A simple form of equalization is Tx de-emphasis as specified in PCIe 1.

55. Is PCIe a protocol or interface?

Legacy PCI is a parallel data transfer protocol. But PCIe is a serial data transfer protocol.

56. What is Type 0 and Type 1 in PCIe?

Type 0 configuration requests are sent only to the device for which they are intended. Type 1 configuration requests are sent to switches/bridges on the way; the last one before the actual target device will convert it to type 0.

57. What is payload size in PCIe?

The PCIe controller will use a maximum data payload size of 2048 bytes. The PCIe controller will use a maximum data payload size of 4096 bytes.

58. What is BDF in PCIe?

BDF stands for the Bus: Device. Function notation used to succinctly describe PCI and PCIe devices. The simple form of the notation is: PCI Bus number in hexadecimal, often padded using a leading zeros to two or four digits.

59. How is PCIe speed calculated?

Maximum PCIe Bandwidth = SPEED * WIDTH * (1 - ENCODING) - 1Gb/s. For example, a gen 3 PCIe device with x8 width will be limited to: Maximum PCIe Bandwidth = 8G * 8 * (1 - 2/130) - 1G = 64G * 0.985 - 1G = ~62Gb/s.

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63. What is the logic used in scrambling?

The logic circuit which performs randomization is known as scrambler. As shown in the figure, it is a generic linear feedback shift register with EX-OR gate. The function of the scrambling is to remove long string of ones (1s) and zeros (0s) from the digital binary data.

64.Why coupling capacitor is used in PCIe?

Coupling capacitors find plenty of uses in analog applications and on differential protocols, acting essentially as high pass filters that remove DC bias carried seen on a signal. In PCIe routing, coupling capacitors are used for the same function (removing DC offset), but for different purposes.